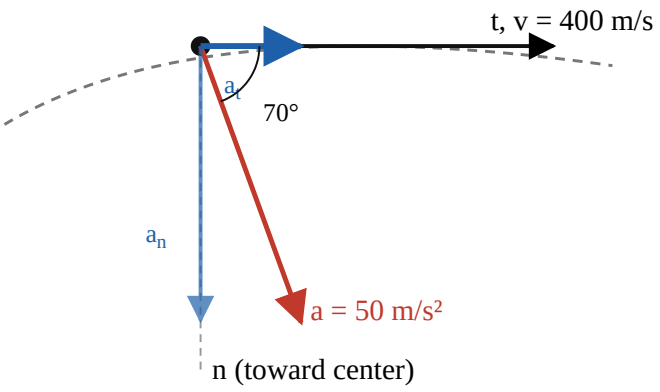
Question 1 **15 points**

CURVILINEAR MOTION – NORMAL & TANGENTIAL COMPONENTS

Given: v = 400 m/s, a = 50 m/s², with a at 70° from the direction of motion (the tangent t), pointing toward the center of curvature.
Find: the rate of increase in speed, $\dot{v} = a_t$ (5 p), and the radius of curvature ρ (10 p).



Rate of increase in speed (tangential component) – 5 p

The rate of change of speed is the tangential component of the acceleration:

$$\dot{v} = a_t = a \cdot \cos 70° = 50 \cdot \cos 70° = 50(0.3420)$$

$$\boxed{\dot{v} = a_t = 17.1 \text{ m/s}^2}$$

Radius of curvature (normal component) – 10 p

$$a_n = a \cdot \sin 70° = 50(0.9397) = 46.98 \text{ m/s}^2$$

Then use $a_n = v^2/\rho$:

$$\rho = v^2 / a_n = (400)^2 / 46.98 = 160\,000 / 46.98$$

$$\boxed{\rho = 3405 \text{ m} \approx 3.41 \text{ km}}$$

Check: $\sqrt{(a_t^2 + a_n^2)} = \sqrt{(17.10^2 + 46.98^2)} = 50.0 \text{ m/s}^2$ ✓

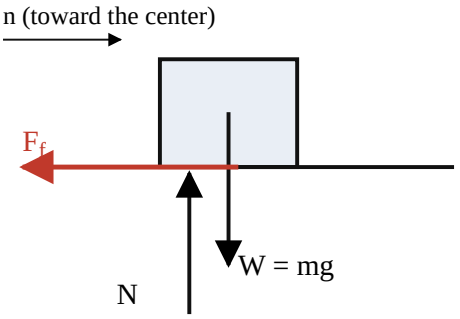
Question 2 — 20 points

EQUATION OF MOTION – NORMAL & TANGENTIAL COORDINATES

Given: m = 1 kg, constant speed v = 4 m/s, $\mu_s = 0.6$, g = 9.81 m/s²

Find: the smallest radius of curvature ρ_{min} so that the cartons do not slip.

Free-body diagram (side view)



Friction is the only horizontal force, so it must supply the centripetal force $m \cdot v^2/\rho$.

Speed is constant ⇒ $a_t = 0$

Equations of motion

$\Sigma F_b = 0$: $N - mg = 0 \Rightarrow N = mg = (1)(9.81) = 9.81$ N

$\Sigma F_t = m \cdot a_t = 0$ (constant speed)

$\Sigma F_n = m \cdot a_n$: $F_f = m \cdot v^2/\rho$

Condition for no slipping

The smallest radius is reached when friction is at its maximum (the carton is about to slip):

$F_f = \mu_s N = (0.6)(9.81) = 5.886$ N

$\mu_s \cdot m \cdot g = m \cdot v^2/\rho_{min} \Rightarrow \rho_{min} = v^2 / (\mu_s g) = (4)^2 / [(0.6)(9.81)] = 16 / 5.886$

$\rho_{min} = 2.72$ m

The mass cancels out, so the answer does not depend on the carton's mass. If $\rho < 2.72$ m, the friction needed would be more than $\mu_s N$ and the carton would slip outward.

Question 3 — 14 points

THEORY

a) Difference between kinematics and kinetics (7 p)

	Kinematics	Kinetics
Definition	Studies the **geometry of motion**: how a body moves, **without considering the forces** that cause the motion.	Studies the **relation between the forces** acting on a body and the **motion they produce**.
Quantities	Position, displacement, velocity, acceleration and time.	Force, mass, moment, momentum, work and energy, together with acceleration.
Main equations	$v = ds/dt$, $a = dv/dt$, $a\,ds = v\,dv$, $a_t = \dot{v}$, $a_n = v^2/\rho$	Newton's 2nd law $\Sigma F = m \cdot a$, work–energy, impulse–momentum
Example	Question 1: finding a_t and ρ from v and a.	Question 2: using friction (a force) to find ρ_{min}.

In short: kinematics answers *"how does it move?"* and kinetics answers *"why does it move that way?"* Kinetics uses the results of kinematics.

b) Conservation of linear momentum (7 p)

The **linear momentum** of a particle is **L = m·v** (a vector, units kg·m/s). The **principle of linear impulse and momentum** says:

$$m \cdot v_1 + \Sigma \int F\,dt = m \cdot v_2$$

If the **sum of the external impulses on a system of particles is zero**, then the total linear momentum of the system stays **constant**:

$$\Sigma\, m \cdot v_1 = \Sigma\, m \cdot v_2 \quad (\text{e.g. } m_A v_{A1} + m_B v_{B1} = m_A v_{A2} + m_B v_{B2})$$

- **Condition:** there are no external impulsive forces on the system, or they are negligible. The **internal forces** between the particles always cancel in pairs (Newton's 3rd law: equal and opposite), so they do not change the total momentum.
- Momentum is a **vector**, so it can be conserved in one direction (e.g. x) even if it is not conserved in another.
- **Applications:** collisions and impact, explosions, gun and bullet recoil, a person jumping off a boat, and coupling of railway cars.
- Note: momentum is conserved in a collision, but **kinetic energy usually is not**, because some of it is lost as heat, sound and deformation.